Lab 4: Test Cases and Testing Results

HDB Resale Flat Search and Analysis System (HomeMatch SG), Team ABCDE

Work product	Author(s)
Black-box test design and `CpfGrantServiceBlackBoxTest`	*fill in*
Basis path test design and the two `*BasisPathTest` classes	*fill in*
Test execution and this report	*fill in*

1. Scope and how to run

Lab 4 item	What we tested	Technique	Test class
3.2.1	Control class `CpfGrantService` (UC-06, UC-07; FR-10 to FR-13)	Equivalence classes + boundary values	`service/CpfGrantServiceBlackBoxTest`
3.2.2 (method 1)	`AuthenticationService.login()` (UC-11; FR-26)	Basis path	`service/AuthenticationServiceBasisPathTest`
3.2.2 (method 2)	`ValuationService.ingest()` (UC-12; FR-29, FR-30)	Basis path	`service/ValuationIngestBasisPathTest`

All tests are automated with JUnit 5, AssertJ and Mockito (Spring Boot test starter). Run them from `backend/`:

```
mvn test                                                    # whole suite (68 tests)
mvn test -Dtest='CpfGrantServiceBlackBoxTest,*BasisPathTest' # only the Lab 4 tests (42 tests)
```

Result of the run on 27 Oct 2026: 68 tests, 0 failures, 0 errors (42 of them are the Lab 4 tests below).

2. Black-box testing of `CpfGrantService`

2.1 Why this class

`CpfGrantService` is the control class behind UC-06 *Calculate CPF Grant Eligibility* and UC-07 *Submit Financial Profile*. It turns a buyer's income, marital status and parents' location into CPF Housing Grant, EHG and PHG amounts, then works out the down payment, loan, monthly instalment and an affordability rating. Getting it wrong tells a buyer they can afford a flat they cannot, so it is the most consequential calculation in the system. It also has many input variables with clear ranges in the SRS, which suits equivalence class and boundary value testing.

Fixture. One 4-room flat in Ang Mo Kio last sold for **$500,000**. Parent postal code `560123` (postal sector 56) is about **1.1 km** away and `640123` (sector 64, Jurong) about **14.3 km** away. Grant rules are the real `grant-rules.yml` (Family Grant S$80,000 for 2- to 4-room up to S$16,000 income, Singles Grant

S$40,000 up to S$8,000; EHG ceiling $9,000; PHG within 4 km; 25 years at 2.6%, 25% minimum down payment, 30% MSR).

Expected values were worked out by hand and with a separate Python script from the SRS and `grant-rules.yml`, not by running the code.

2.2 Equivalence classes

Input (source)	Valid classes	Invalid classes
Monthly household income (FR-10)	V1 family: 0 to 9,000 (EHG band applies) V2 family: 9,001 to 1,000,000 (no EHG) V3 single: 0 to 4,500 (half EHG at 2x income) V4 single: 4,501 to 1,000,000 (no EHG)	I1 below 0 I2 above 1,000,000 I3 missing
Income for the CPF Housing Grant (FR-11)	V21 family: up to 16,000 V22 family: above 16,000 (no grant) (singles: up to 8,000 / above 8,000)	
Flat type (FR-11)	V23 2- to 4-room (larger grant) V24 5-room, Executive, Multi-generation (smaller grant)	
Marital status (FR-10)	V5 SINGLE V6 MARRIED or ENGAGED (family nucleus)	I4 missing
Living with parents (FR-11)	V7 yes V8 no	none (boolean)
Parents' postal code (UC-07)	V9 blank (no PHG check) V10 known sector within 4 km V11 known sector beyond 4 km	I5 not 6 digits I6 6 digits but unknown sector
Loan tenure, years (FR-13)	V12 5 to 30 V13 not given (default 25)	I7 below 5 I8 above 30
Down payment % (FR-13)	V14 0 to under 25 (raised to 25 with a notice) V15 25 to 100 (used as given) V16 not given (default 25)	I9 below 0 I10 above 100
Interest rate % (FR-13)	V17 0 (zero-interest formula) V18 above 0 to 10 V19 not given (default 2.6)	I11 below 0 I12 above 10
Flat	V20 exists	I13 unknown id
Affordability (output, FR-12)	O1 instalment/income at most 25% COMFORTABLE O2 above 25% to 30% STRETCHED O3 above 30% ABOVE_LIMIT O4 income 0 UNKNOWN	

2.3 Boundary values

Boundary	Values tested	Test ids
EHG first band edge (family)	1,500 / 1,501	BB-01, BB-02
EHG ceiling (family)	9,000 / 9,001	BB-03, BB-04
EHG ceiling (single, 2x income)	4,500 / 4,501	BB-05, BB-06
CPF Housing Grant ceiling	family 16,000 / 16,001; single 8,000 / 8,001	BB-28
Income lower limit	-1 / 0	BB-12, BB-07 / BB-21
Income upper limit	1,000,000 / 1,000,001	BB-21, BB-13
Tenure	4 / 5 / 30 / 31	BB-14, BB-08, BB-09, BB-15
Down payment	-0.01 / 0 / 24.99 / 25 / 100 / 100.01	BB-16, BB-08, BB-10, BB-11, BB-09, BB-17
Interest	-0.01 / 0 / 10 / 10.01	BB-18, BB-08, BB-09, BB-19
PHG distance	4.0 km / 4.01 km	BB-26
Affordability ratio	25.0% / 25.1% / 30.0% / 30.1%	BB-27
BB-28	V21, V22, V23, V24 boundaries	Married 16,000 / 16,001 (4-room); Single 8,000 / 8,001 (4-room); Engaged 5,000 (5-room); Single 5,000 (Executive); Married 5,000 (2-room)

Keeping the number of cases down. Each valid test covers several valid classes at once (for example BB-08 covers V12 lower edge, V14 lower edge and V17 together). Each invalid test changes exactly one input away from a valid baseline, so a rejection can only come from that input. Boundaries were chosen at the edges where behaviour changes in the SRS (band edges, ceilings, the 25% minimum, the 4 km rule, the MSR thresholds) rather than at arbitrary points.

2.4 Test cases and results

Unless stated, loan fields are left blank (defaults: 25 years, 2.6%, 25% down) and the user is not living with parents. “Rating” is the affordability rating. All amounts in S$.

Id	Classes	Test input	Expected output	Actual output
BB-01	V1, V6, V8, V9, V13, V16, V19, V21, V23, O3	Married, income 1,500, no postal code	Family Grant 80,000; EHG 120,000; PHG 0 (“Add your parents’ postal code”); down 125,000; loan 175,000; instalment 794; ABOVE_LIMIT; defaults 25 y / 2.6% / 25% shown	Same. Pass
BB-02	V1 (edge)	Married, income 1,501	Family Grant 80,000; EHG 110,000; loan 185,000; instalment 839	Same. Pass

Id	Classes	Test input	Expected output	Actual output
BB-03	V1 (edge), O1	Married, income 9,000	Family Grant 80,000; EHG 5,000; loan 290,000; instalment 1,316; COMFORTABLE	Same. Pass
BB-04	V2 (edge), V6, V10	Engaged, income 9,001, parents at 560123	Family Grant 80,000; EHG 0; PHG 20,000; parent distance 1.1 km; instalment 1,248; COMFORTABLE	Same. Pass
BB-05	V3 (edge), V5, V7	Single, income 4,500, living with parents	Singles Grant 40,000; EHG 2,500; PHG 15,000; loan 317,500; instalment 1,440	Same. Pass
BB-06	V4 (edge)	Single, income 4,501	Singles Grant 40,000; EHG 0; PHG 0; instalment 1,520	Same. Pass
BB-07	V1 (edge 0), V11, O4	Married, income 0, parents at 640123	Family Grant 80,000; EHG 120,000; PHG 0 (“more than 4 km”); distance 14.3 km; rating UNKNOWN	Same. Pass
BB-08	V12, V14, V17 (lower edges)	Married, 9,001; tenure 5; down 0%; interest 0%	Down raised to 25% (125,000) with a notice; grants 80,000; loan 295,000; instalment 4,917 (295,000 / 60)	Same. Pass
BB-09	V12, V15, V18 (upper edges)	Married, 9,001; tenure 30; down 100%; interest 10%	Down 500,000; loan 0; instalment 0; COMFORTABLE	Same. Pass
BB-10	V14 (upper edge)	Married, 9,001; down 24.99%	Raised to 25% with notice; instalment 1,338	Same. Pass
BB-11	V15 (lower edge)	Married, 9,001; down 25%	25% used, no notice; instalment 1,338	Same. Pass
BB-12	I1	Income -1	Rejected on `profile.householdIncome` (HTTP 422)	Same. Pass
BB-13	I2	Income 1,000,001	Rejected on `profile.householdIncome`	Same. Pass
BB-14	I7	Tenure 4	Rejected on `loanTenureYears`	Same. Pass
BB-15	I8	Tenure 31	Rejected on `loanTenureYears`	Same. Pass
BB-16	I9	Down -0.01%	Rejected on `downPaymentPercent`	Same. Pass
BB-17	I10	Down 100.01%	Rejected on `downPaymentPercent`	Same. Pass
BB-18	I11	Interest -0.01%	Rejected on `interestRatePercent`	Same. Pass
BB-19	I12	Interest 10.01%	Rejected on `interestRatePercent`	Same. Pass

Id	Classes	Test input	Expected output	Actual output
BB-20	I5	Postal code `12345`	Rejected on `profile.parentPostalCode`	Same. Pass
BB-21	valid edges	(a) income 0, tenure 5, down 0, rate 0; (b) income 1,000,000, tenure 30, down 100, rate 10	No validation errors	Same. Pass
BB-22	I6	Postal code `990000`	422, field error on `parentPostalCode` (UC-07.EX.2)	Same. Pass
BB-23	I4	Marital status missing, no saved profile	422 `PROFILE_REQUIRED` (UC-06.EX.1)	Same. Pass
BB-24	I3	Income missing, no saved profile	422 `PROFILE_REQUIRED`	Same. Pass
BB-25	I13	Flat id 999	404 `NOT_FOUND`	Same. Pass
BB-26	PHG boundary	Married, not with parents, distance 4.0 km / 4.01 km	20,000 / 0	Same. Pass
BB-27	O1, O2, O3 boundaries	Income 10,000; instalment 2,500 / 2,510 / 3,000 / 3,010	COMFORTABLE / STRETCHED / STRETCHED / ABOVE_LIMIT	Same. Pass

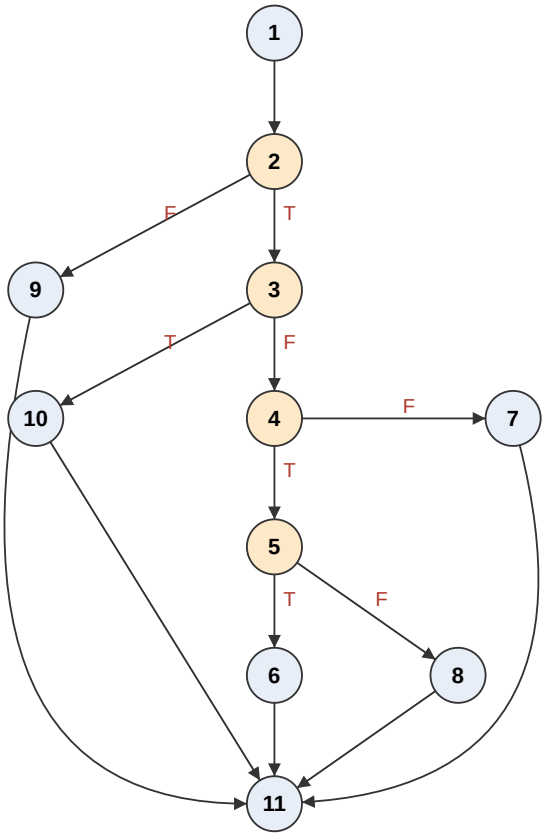
BB-26 to BB-28 call `phg()`, `affordability()` and `cpfHousingGrant()` directly, because no real postal sector sits exactly 4 km from a flat and no real loan gives an instalment of exactly 25% of income.

3. White-box (basis path) testing

3.1 Method 1: `AuthenticationService.login()`

Implements UC-11 with the lockout rule (5 failed attempts lock the account for 15 minutes, TBD-4).

Flow graph: AuthenticationService.login()



Flow graph of login()

Node	Code (`AuthenticationService.java`)
1	Look up the user by trimmed email (lines 91 to 93)
2	**P1** user found?
3	**P2** `user.isLocked(now)` (line 94)
4	**P3** password does not match (line 97)
5	`failures = attempts + 1`; **P4** `failures >= maxFailedLogins` (lines 98, 99)
6	Reset count, lock for 15 minutes, save, throw 423 (lines 100 to 103)
7	Reset count and lock, save, start session, return token (lines 109 to 112)
8	Save the new count, throw 401 (lines 105 to 107)
9	Throw 401 `INVALID_CREDENTIALS` (line 93)
10	Throw 423 `ACCOUNT_LOCKED` (line 95)
11	Exit

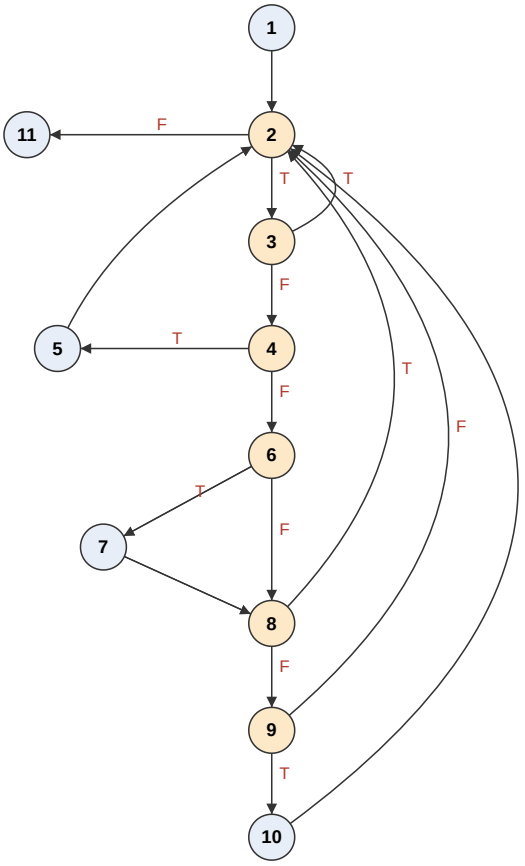
Cyclomatic complexity. Edges E = 14, nodes N = 11, so V(G) = E - N + 2 = **5**. Check: 4 predicate nodes + 1 = 5, and the graph has 5 regions.

Id	Basis path	Test input	Expected output	Actual output
WB-A1	1-2-9-11	Email `nobody@example.com` (not registered)	401 `INVALID_CREDENTIALS`; nothing saved	Same. Pass
WB-A2	1-2-3-10-11	Registered user locked for 10 more minutes; correct password	423 “locked for 10 more minutes”; password not even checked	Same. Pass
WB-A3	1-2-3-4-5-8-11	Registered user, 0 previous failures, wrong password	401; failed attempts now 1; not locked	Same. Pass
WB-A4	1-2-3-4-5-6-11	Registered user, 4 previous failures, wrong password	423 “locked for 15 more minutes”; locked until now + 15 min; count reset to 0	Same. Pass
WB-A5	1-2-3-4-7-11	Email with spaces around it, 3 previous failures, expired lock, correct password	Token returned; count 0; lock cleared; session saved	Same. Pass

3.2 Method 2: `ValuationService.ingest()`

Part of the 24-hour batch (UC-12). It takes new resale records from data.gov.sg, skips records with no town, flags towns with incomplete records so their baseline is not updated (UC-12.EX.2), matches each record to a known flat or geocodes a new one, and stores transactions it has not seen before.

Flow graph: ValuationService.ingest()



Flow graph of ingest()

Node	Code (`ValuationService.java`)
1	Set up result set, town map and counter (lines 198 to 201)
2	**P1** another record in the list? (line 203)
3	**P2** town missing or blank? (line 204)
4	**P3** price or area not positive, or month or flat type missing? (line 207)
5	Flag the town as incomplete (line 208)
6	Look up the flat; **P4** not found? (lines 211 to 213)
7	`createFlat()`: geocode and save a new flat (line 214)
8	**P5** still no flat? (line 216)
9	**P6** transaction not stored yet? (line 220)
10	Save transaction, update the flat's latest price, count it (lines 221 to 224)
11	Return the result (line 228)

Cyclomatic complexity. E = 16, N = 11, so V(G) = 16 - 11 + 2 = **7**. Check: 6 predicate nodes + 1 = 7.

The basis set was built with the baseline method: start from the most common path (WB-B4, a new sale of a known flat), then flip one decision at a time.

Id	Basis path	Test input	Expected output	Actual output
WB-B1	1-2-11	Empty list	0 inserted; no towns flagged	Same. Pass
WB-B2	1-2-3-2-11	One record with town `" "`	0 inserted; no towns flagged; nothing saved	Same. Pass
WB-B3	1-2-3-4-5-2-11	One Ang Mo Kio record with price 0	0 inserted; ANG MO KIO flagged incomplete	Same. Pass
WB-B4	1-2-3-4-6-8-9-10-2-11	New sale (Sep 2026, $510,000) of a known flat last sold for $480,000	1 inserted; flat's latest price now 510,000 (2026-09); no geocoding call	Same. Pass
WB-B5	1-2-3-4-6-8-9-2-11	Same sale, already stored	0 inserted; latest price stays 480,000	Same. Pass
WB-B6	1-2-3-4-6-7-8-9-10-2-11	Sale of unknown block 999; geocoder returns an exact point	New flat created; 1 inserted	Same. Pass
WB-B7	1-2-3-4-6-7-8-2-11	Sale of unknown block 999; geocoder only returns an approximate point	Skipped: 0 inserted; no flat saved	Same. Pass

3.3 Evidence that the tests do their job

Coverage (JaCoCo 0.8.12, Lab 4 tests only). `login()`: 100% of instructions and branches. `ingest()`: 100% of instructions and 80% of branches. The 4 branches not taken are single conditions inside the two compound predicates (P2 has 2 conditions, P3 has 4). Basis path testing treats each compound predicate as one decision, as in the lecture method, so this is expected. Counting every simple condition as its own node would give V(G) = 11, which is also what JaCoCo reports.

Fault injection. We introduced four realistic one-character bugs and ran the Lab 4 tests against each. Every bug was caught, then the code was restored.

Bug introduced	Caught by
PHG: `parentKm <= 4` changed to `< 4`	BB-26
Affordability: `ratio <= 25` changed to `< 25`	BB-27
Lockout: `failures >= 5` changed to `> 5` (locks on the 6th try)	WB-A4
Ingest: duplicate check removed	WB-B5

4. Traceability

Requirement	Use case	Design class	Code	Test cases
FR-10 financial profile input	UC-07	`CpfGrantService`, `FinancialProfileDto`	`estimate()`, `saveProfile()`	BB-12, BB-13, BB-20 to BB-24
FR-11 CPF Housing Grant, EHG and PHG estimate	UC-06	`CpfGrantService`, `GrantRules`	`cpfHousingGrant()`, `ehg()`, `phg()`	BB-01 to BB-07, BB-26, BB-28
FR-12 instalment and assumptions	UC-06	`CpfGrantService`	`estimate()`, `monthlyInstalment()`, `affordability()`	BB-01, BB-08, BB-09, BB-27
FR-13 adjustable loan assumptions	UC-06	`CpfGrantRequestDto`	Bean Validation constraints	BB-08 to BB-11, BB-14 to BB-19, BB-21
FR-26 lockout after failed logins	UC-11	`AuthenticationService`	`login()`	WB-A1 to WB-A5
FR-29 ingest new transactions	UC-12	`ValuationService`, `ExternalApiGateway`	`ingest()`	WB-B1, WB-B4 to WB-B7
FR-30 / UC-12.EX.2 incomplete data	UC-12	`ValuationService`	`ingest()`	WB-B2, WB-B3

5. Limitations

- Tests replace the database and external APIs with Mockito mocks, so they check the logic, not the SQL or the real government APIs. `ApiFlowTest` covers the full HTTP stack against an embedded H2 database.
- Grant amounts are tested against `grant-rules.yml`, so the file itself was checked on 27 Oct 2026. The EHG table for families, HDB's own EHG table for singles and the PHG amounts all match. The

CPF Housing Grant (Family Grant / Singles Grant) for resale flats was missing, so it was added on 27 Oct 2026 (BB-01 to BB-11 and BB-28 cover it). FR-11 in the SRS should be updated to name it.

- **Finding: the screen and the API disagree on loan limits.** The CPF panel's sliders allow tenure 5 to 25 years, down payment 25% to 60% and interest 1% to 5%, but the API accepts tenure up to 30 years, down payment 0% to 100% and interest 0% to 10% (BB-08, BB-09, BB-21 pass at the API). Nothing breaks, but the two should match. If the team decides HDB loans stop at 25 years, change `@Max(30)` in `CpfGrantRequestDto` to 25 and update BB-09, BB-15 and BB-21.